

Continuum Mechanics HW1

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Technion

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1 Divergence of a Tensor in Cylindrical Coordinates

In the tutorial we saw the definition of the divergence of a tensor:

$$\nabla \cdot \tau = \mathbf{e}_i \cdot \frac{\partial}{\partial x_i} (\tau_{kl} \mathbf{e}_k \mathbf{e}_l) \quad (1)$$

prove that in cylindrical Coordinates

$$\nabla \cdot \tau = \begin{pmatrix} \frac{\partial \tau_{rr}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta r}}{\partial \theta} + \frac{\partial \tau_{zr}}{\partial z} + \frac{\tau_{rr} - \tau_{\theta\theta}}{r} \\ \frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{z\theta}}{\partial z} + \frac{\tau_{r\theta} + \tau_{\theta r}}{r} \\ \frac{\partial \tau_{rz}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \tau_{zz}}{\partial z} + \frac{\tau_{rz}}{r} \end{pmatrix} \quad (2)$$

Starting from the definition and developing for cylindrical coordinates explicitly

$$\nabla \cdot \tau = \mathbf{e}_i \cdot \frac{\partial}{\partial x_i} (\tau_{kl} \mathbf{e}_k \mathbf{e}_l) \quad (3)$$

$$= \left(\hat{\mathbf{r}} \cdot \frac{\partial}{\partial r} + \hat{\theta} \cdot \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{z}} \cdot \frac{\partial}{\partial z} \right) (\tau_{ij} \mathbf{e}_i \mathbf{e}_j) \quad (4)$$

$$= \left(\hat{\mathbf{r}} \cdot \frac{\partial}{\partial r} + \hat{\theta} \cdot \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{z}} \cdot \frac{\partial}{\partial z} \right) (\tau_{ij}) \mathbf{e}_i \mathbf{e}_j \quad (5)$$

$$+ \left(\hat{\mathbf{r}} \cdot \frac{\partial}{\partial r} + \hat{\theta} \cdot \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{z}} \cdot \frac{\partial}{\partial z} \right) (\mathbf{e}_i) \tau_{ij} \mathbf{e}_j \quad (6)$$

$$+ \left(\hat{\mathbf{r}} \cdot \frac{\partial}{\partial r} + \hat{\theta} \cdot \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\mathbf{z}} \cdot \frac{\partial}{\partial z} \right) (\mathbf{e}_j) \tau_{ij} \mathbf{e}_i \quad (7)$$

$$= \left(\frac{\partial \tau_{ri}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta i}}{\partial \theta} + \frac{\partial \tau_{zi}}{\partial z} \right) \mathbf{e}_i \quad (8)$$

$$+ \left(\frac{\tau_{ri}}{r} \right) \mathbf{e}_i \quad (9)$$

$$+ \left(\frac{\tau_{\theta r}}{r} \hat{\theta} - \frac{\tau_{\theta\theta}}{r} \hat{\mathbf{r}} \right) \quad (10)$$

Equation 8 follows from 5 because the operator acts on the components of the tensor itself and the unit vectors then give deltas giving the resulting form, then the remaining index is renamed to i. Equation 9 follow from 6 because in the basis vectors of cylindrical coordinates, the vectors only depend on theta so derivatives with respect to other parameters are zero. Equation 10 follows from 7 for the same reason as 9. We can now rearrange this and find

$$\nabla \cdot \tau = \begin{pmatrix} \frac{\partial \tau_{rr}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta r}}{\partial \theta} + \frac{\partial \tau_{zr}}{\partial z} + \frac{\tau_{rr} - \tau_{\theta\theta}}{r} \\ \frac{\partial \tau_{r\theta}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta\theta}}{\partial \theta} + \frac{\partial \tau_{z\theta}}{\partial z} + \frac{\tau_{r\theta} + \tau_{\theta r}}{r} \\ \frac{\partial \tau_{rz}}{\partial r} + \frac{1}{r} \frac{\partial \tau_{\theta z}}{\partial \theta} + \frac{\partial \tau_{zz}}{\partial z} + \frac{\tau_{rz}}{r} \end{pmatrix} \quad (11)$$

2 Heat Diffusion

1. What are the dimensional quantities in the problem? What are their dimensions? Separate your answer into parameters, dependent variables, and independent variables.
2. Find 2 dimensionless combinations. Using them, write a self-similar solution, as we saw in the tutorial. Does your solution satisfy the boundary conditions?
3. Substitute the solution into the heat equation to obtain a second-order ordinary differential equation.
4. Solve the equation. What is the boundary condition at $x \rightarrow \infty$?
5. Plot the temperature distribution along the rod at different times?

2.1 Dimensions

The dimensional parameter in this problem are k the thermal diffusion coefficient with dimensions $L^2 t^{-1}$ and T_0 which has dimensions of TL where T is temperature and L is length. The independent variables are x which has dimensions of length and t which has dimension of time. The dependent variable in the problem is the temperature at each point, which has dimensions of temperature.

2.2 Dimensionless combinations

Similarly to the training we can define

$$\xi = \frac{x}{\sqrt{kt}} = x (kt)^{-1/2} \qquad \theta = \frac{T}{T_0} \quad (12)$$

Which then gives us the self-similar solution

$$\theta = \Phi(\xi) \rightarrow T = T_0 \Phi(\xi) \quad (13)$$

We require new boundary conditions:

$$\Phi(0) = 1 \qquad \Phi(x \rightarrow \infty) = 0 \quad (14)$$

2.3 Inserting Solution

Rewriting the left hand side first:

$$\partial_t T = T_0 \partial_t \Phi(\xi) = T_0 \frac{\partial \Phi}{\partial \xi} \frac{\partial \xi}{\partial t} = -\frac{x T_0}{2 k^{-1/2} t^{-3/2}} \frac{\partial \Phi}{\partial \xi} = -\frac{\xi T_0}{2 t} \frac{\partial \Phi}{\partial \xi} \quad (15)$$

$$\partial_x T = T_0 \partial_x \Phi = T_0 \frac{\partial \Phi}{\partial \xi} \frac{\partial \xi}{\partial x} = \frac{T_0}{\sqrt{kt}} \frac{\partial \Phi}{\partial \xi} \quad (16)$$

therefore the equation becomes

$$-\frac{\xi T_0}{2 t} \frac{\partial \Phi}{\partial \xi} = \frac{k T_0}{kt} \frac{\partial^2 \Phi}{\partial \xi^2} \quad (17)$$

rearranging to

$$\frac{\partial^2 \Phi}{\partial \xi^2} + \frac{\xi}{2} \frac{\partial \Phi}{\partial \xi} = 0 \quad (18)$$

which is indeed a second order ordinary differential equation.

2.4 Solving

Using the hint the solution is

$$\Phi = A \operatorname{erf}\left(\frac{\xi}{2}\right) + B \quad (19)$$

Using the boundary condition that $\Phi(\xi = 0) = 1$ we find that $B = 1$. Using the boundary condition that $\Phi(x \rightarrow \infty) = 0$ we find that $A = -1$. Therefore the solution is

$$\Phi(\xi) = 1 - \operatorname{erf}\left(\frac{\xi}{2}\right) \quad (20)$$

And the original equation's solution is

$$T = T_0 \left(1 - \operatorname{erf}\left(\frac{\xi}{2}\right)\right) \quad (21)$$

2.5 Plot

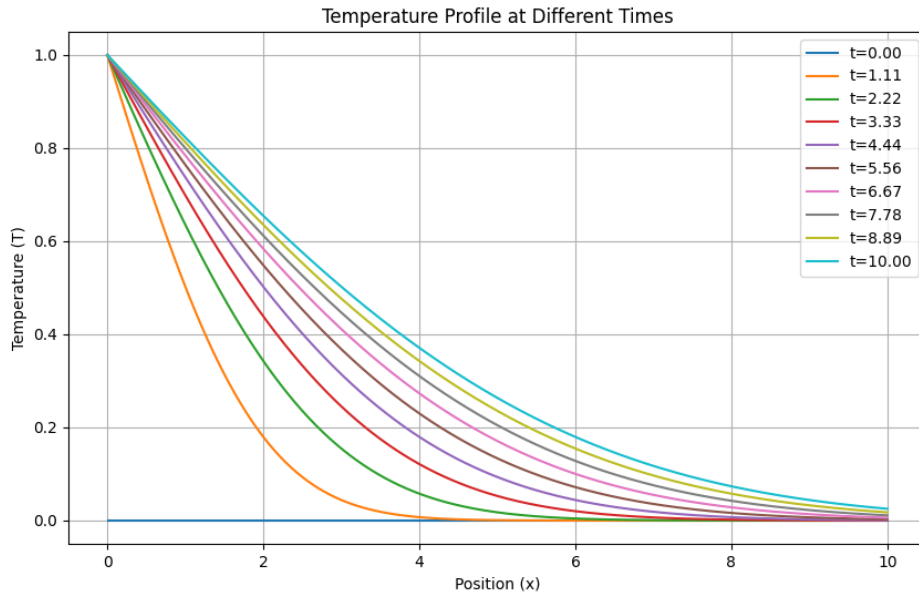


Figure 1 – The temperature profile along the x axis of the rod at different times, as given by equation 21.

Listing 1 – The code

```

import matplotlib.pyplot as plt
import numpy as np
from scipy.special import erf

def plot_profile_at_times(times, x, T):
    plt.figure(figsize=(10, 6))
    for t in times:
        plt.plot(x, T(x, t), label=f't={t:.2f}')
    plt.xlabel('Position_□(x)')
    plt.ylabel('Temperature_□(T)')
    plt.title('Temperature_□Profile_□at_□Different_□Times')
    plt.legend()
    plt.grid()
    plt.show()

def main():
    # temperature and coefficient are both assumed to be 1 for simplicity
    # the position also isn't really the real position, scaling will be off
    # but I imagine it's not very important for this.
    held_temp = 1
    xi = lambda x, t: x/np.sqrt(t)
    solution = lambda x, t: held_temp*(1-erf(xi(x, t)/2))
    times = np.linspace(0, 10, 10)
    x = np.linspace(0, 10, 1000)
    plot_profile_at_times(times, x, solution)

if __name__ == "__main__":
    main()

```

3 Fluid with Self-Gravity

Consider a fluid with a constant density that is in equilibrium with its own gravitational force $\rho \mathbf{f} = -\rho \nabla \phi$, where the potential ϕ satisfies Poisson's equation $\nabla^2 \phi = 4\pi G \rho$ and G is Newton's gravitational constant. Assume spherical symmetry around the origin and find the pressure $p = p(r)$ up to an additive constant.

Since the liquid is in hydrostatic equilibrium and $\rho \mathbf{f}$ is a gradient of a scalar potential, the following is true

$$\rho \mathbf{f} = \nabla p \quad (22)$$

we can then insert this into the equation for the force

$$\nabla \phi = -\frac{\nabla p}{\rho} \quad (23)$$

inserting this into Poisson's equation

$$\nabla^2 p = -4\pi G \rho^2 \quad (24)$$

To continue in spherical we use the laplacian in spherical coordinates

$$\nabla^2 f = \frac{1}{r} \frac{\partial^2}{\partial r^2} (r f) \quad (25)$$

where the terms of the laplacian which are derivatives of non-r coordinates are ignored because of the spherical symmetry. Therefore the equation is

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial p}{\partial r} \right) = -4\pi G \rho^2 \quad (26)$$

Since the right hand side doesn't depend on r we can integrate once and get

$$r^2 \frac{\partial p}{\partial r} = -\frac{4}{3} \pi G \rho^2 r^3 + A \quad (27)$$

therefore our candidate solution is

$$p = -\frac{2}{3} \pi G \rho^2 r^2 - \frac{A}{r} + B \quad (28)$$

and since we must satisfy equation 22

$$\nabla p = \frac{A}{r^2} - \frac{4}{3} \pi G \rho^2 r = \rho \mathbf{f} \quad (29)$$

Since the force must remain finite even at the origin, $A = 0$. Therefore

$$p = B - \frac{2}{3} \pi G \rho^2 r^2 \quad (30)$$

4 Bouyance force - hot air balloon

1. Write the atmospheric density profile $\rho(z)$ as a function of the height z .
2. We want to treat the hot air balloon as a point particle; to do so, we want to ensure that density changes are small relative to the size of the hot air balloon. This can be required by the condition $\frac{R}{\rho(z)} \frac{\partial \rho}{\partial z} \ll 1$. What is the resulting condition on the balloon's radius R ? Is it satisfied for the problem's parameters?
3. Treat the balloon as a point particle at height z (meaning the air density around it is $\rho(z)$) and write the net force acting on the balloon together with the basket. What is the condition for the balloon to remain at a certain height? Find the height z to which the hot air balloon reaches.

4.1 Density Profile

Since we assume the atmosphere is an ideal gas,

$$pV = Nk_B T \implies p = \frac{N}{V} k_B T \quad (31)$$

but

$$\frac{N}{V} = n = \frac{\rho}{m} \quad (32)$$

therefore

$$p = \frac{\rho k_B T}{m} \quad (33)$$

inserting into the hydrostatic equation

$$\nabla p = \rho \mathbf{f} \quad (34)$$

we get

$$\frac{k_B T}{m} \frac{d\rho}{dz} = -\rho g \quad (35)$$

which is an ODE with the solution

$$\rho(z) = \rho(0) e^{-\frac{z}{l_a}} \quad (36)$$

4.2 Point Balloon

We require that

$$\frac{R}{\rho(z)} \frac{\partial \rho}{\partial z} \ll 1 \implies \frac{d\rho}{dz} \ll \frac{\rho(z)}{R} \quad (37)$$

if we take the derivative of the density profile we found

$$\frac{d\rho}{dz} = -\frac{\rho(0)}{l_a} e^{-\frac{z}{l_a}} \quad (38)$$

and inserting the profile we find

$$-\frac{\rho(0)}{l_a} e^{-\frac{z}{l_a}} \ll \frac{\rho(0) e^{-\frac{z}{l_a}}}{R} \quad (39)$$

simplifying

$$-\frac{1}{l_a} \ll \frac{1}{R} \quad (40)$$

therefore

$$l_a \gg R \quad (41)$$

in our question R is 3 orders of magnitude smaller than l_a therefore the condition is satisfied.

4.3 Constant Height Condition

The balloon is affected by the following forces: its own gravitational pull, the basket's gravitational pull, and the bouyancy force from the balloon. The balloon will reach the height at which it is at equilibrium:

$$\rho_{hot}Vg = \rho(z)Vg + Mg \quad (42)$$

rearranging

$$\rho(z) = \frac{\rho_{hot}Vg + Mg}{Vg} = \frac{\rho_{hot}V + M}{V} \quad (43)$$

Inserting the expression for the density

$$e^{-\frac{z}{l_a}} = \frac{\rho_{hot}V + M}{\rho(0)V} \quad (44)$$

therefore

$$z = -l_a \ln \frac{\rho_{hot}V + M}{\rho(0)V} \quad (45)$$

inserting the numbers we find

$$z = \dots = 1.595 \text{ km} \quad (46)$$